

MUHAMMAD SAMMAR ISLAM

Chakwal, Pakistan

+966-551937463 samarislam678@gmail.com LinkedIn | Transferable Iqama

EDUCATION

Ghulam Ishaq Khan Institute (GIKI), Pakistan

BSc - Electrical and Electronics Engineering — CGPA: 2.90/4.00

2021 – 2025

Swabi, Pakistan

Professional Summary

Motivated Electrical and Electronics Engineering undergraduate from GIKI with hands-on experience in industrial automation, circuit design, IoT systems, and PCB prototyping. Proficient in Python, MATLAB, and embedded platforms such as Arduino and PIC microcontrollers. Passionate about developing intelligent hardware solutions through innovation and problem-solving.

EXPERIENCE

Intern at Oil and Gas Field (POL)

June 2023 – July 2023

Role — Project Engineer

Khaur, Pakistan

- Gained hands-on understanding of power generation, transmission, and POL's 3.2 MW independent power system.
- Explored the operation and maintenance of electrical systems in industrial environments.
- Learned about instrumentation systems, focusing on industrial sensors and their real-world applications.

Intern at Pakistan Aeronautical Complex (PAC)

July 2024

Role — Design Engineer

Kamra, Pakistan

- Gained hands-on aerospace experience by 80% working on aircraft and avionics design projects.
- Enhanced design and testing efficiency by 70% through development and evaluation of avionics and aircraft systems.
- Assisted engineers in troubleshooting and maintaining avionics hardware and control systems.

Intern at Creative Electronics and Automation (Crest)

July 2024 – August 2024

Role — Product Design Engineer

Rawalpindi, Pakistan

- Worked with Pakistan's only fully automated LED manufacturing and testing facility.
- Learned about automation systems, industrial IT solutions, and electronic product design.
- Supported the development of automation products and LED assemblies using modern tools.

PROJECTS

Rotary Inverted Pendulum (Quanser Qube-Servo) | PID Control, MATLAB, Simulink

Jan 2025

- Designed and implemented a rotary inverted pendulum system inspired by the Quanser Qube Servo 3, achieving stable control within 3 seconds.
- Developed and validated a state-space model, improving system accuracy by 85%.
- Implemented and tuned PID controllers in MATLAB/Simulink, reducing oscillations by 70%.

Starter for Three-Phase Induction Motor | Star-Delta, TPDT

Sep 2024

- Designed and developed a motor starter delivering optimal torque and smooth startup performance.
- Implemented current control and overload protection, reducing startup current by 40% and enhancing motor lifespan.

Active Noise Cancellation System (ANC) | Communication, Signal, Python

Feb 2023

- Designed and evaluated an active noise cancellation system for headphones achieving high noise reduction.
- Implemented algorithms to improve sound quality and reduce interference using Python signal processing tools.

TECHNICAL SKILLS

Languages: Python, C++, Assembly, MATLAB, PLC, LaTeX

Technologies/Frameworks: Altium Designer, AutoCAD, DSP, DLD

Developer Tools: CREO, MPLAB, LTSpice, PSpice, LabVIEW, Multisim

EXTRACURRICULAR

- Member of “Institution of Engineering and Technology” (IET).
- Head of SUMMIT event, IET chapter.
- Organized “Student Gala” (2020) and “Science Fair”.
- Member of “GIKI Sports Society” (GSS).
- Winner of Inter-Campus Sports Event (2019–2020).